

Forecasting seasonal changepoint models utilizing changepoint uncertainties

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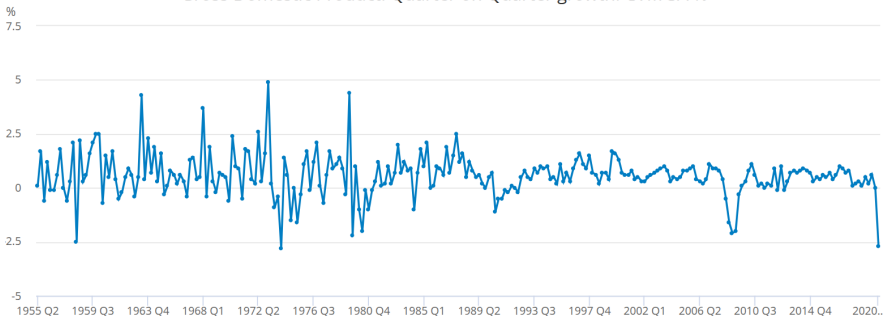
Joint work with Vasilis Pavlopoulos, E. Can, H. Pham, P. Bhatt (Alabama, Huntsville)

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Usual approach

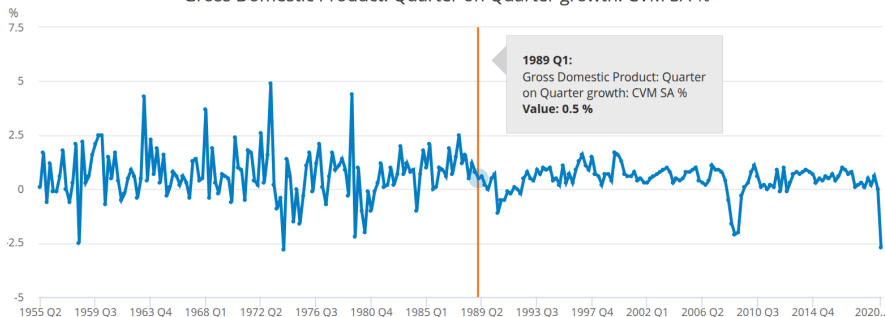


Gross Domestic Product: Quarter on Quarter growth: CVM SA %

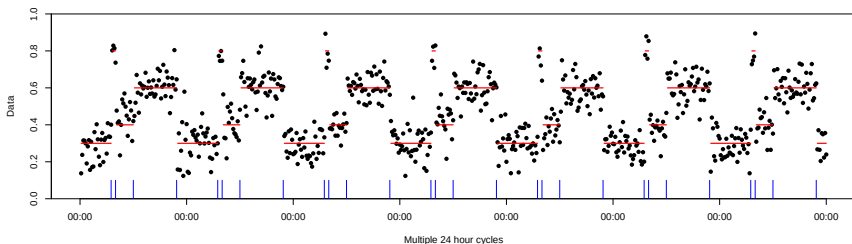


<https://doi.org/10.1002/qre.2712>

Gross Domestic Product: Quarter on Quarter growth: CVM SA %



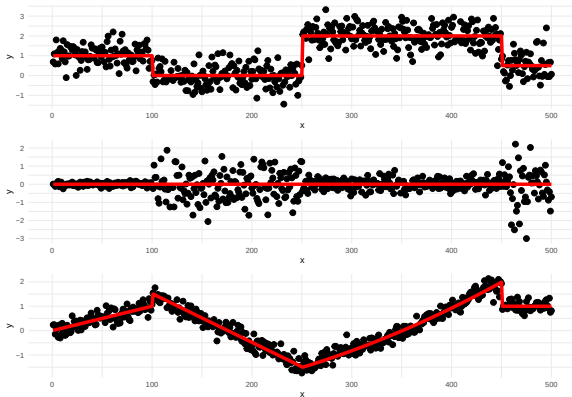
Within Season?



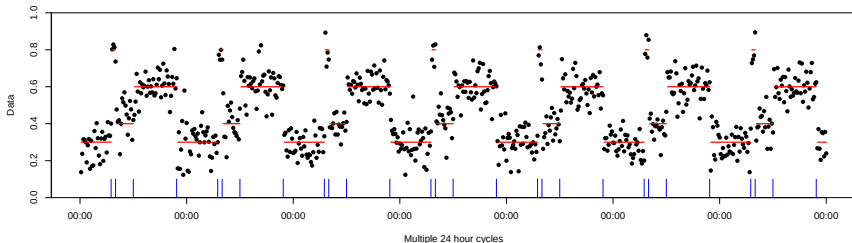
- What are changepoints?
- Within Season changepoint models
- Confidence sets for changepoint problems
- Forecasting with confidence sets
- Back to our examples

For data $y_1, \dots, y_n$, if a changepoint exists at τ , then $y_1, \dots, y_\tau$ differ from $y_{\tau+1}, \dots, y_n$ in some way.

There are many different types of change.

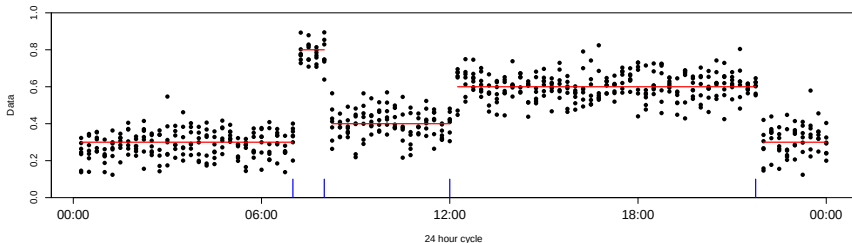


Time is linear





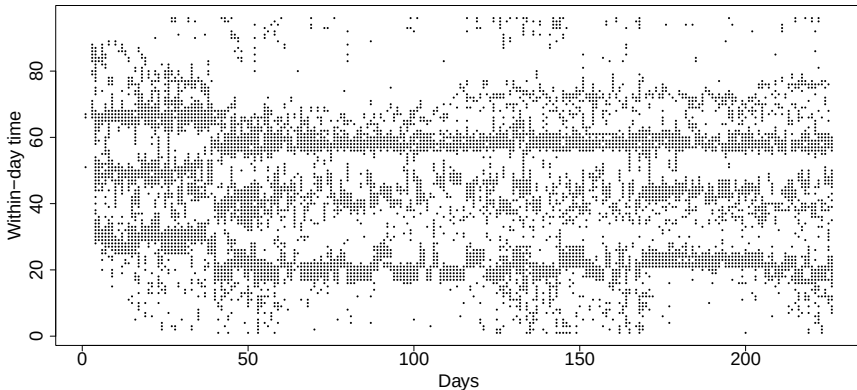
Time is circular





1 Season = 1 series

Howz Data

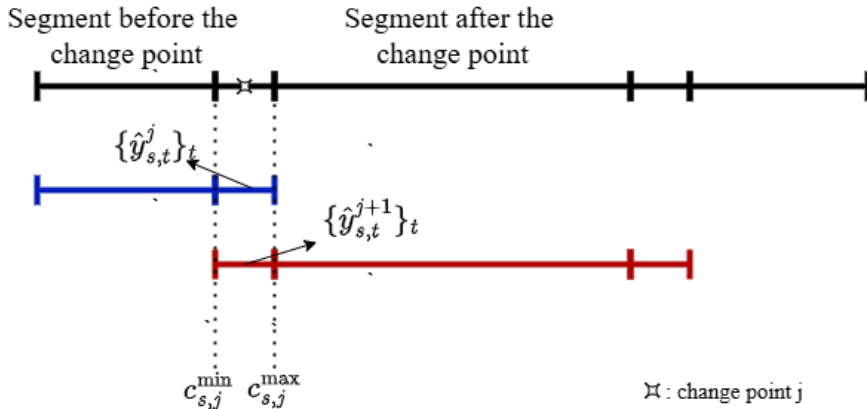


Can analyse with univariate or multivariate techniques.

- Forecasting method is agnostic as to **how** you find the changepoints
- Just use an appropriate confidence set to match your changepoint detection
- If you use univariate techniques you have an extra averaging step for combining the forecasts

Given a cpt detection and confidence set pair, we

1. Draw a realisation from the model for each segment: extending time from the start of the previous confidence set, to the end of the following confidence set.
2. Blend the forecasts for the confidence set period using an appropriate monotonic weighting (cumulative likelihood).
3. If univariate, then combine across forecasts.



To forecast season, s , at time, t , we

$$\hat{y}_{s,t} = \begin{cases} \hat{y}_{s,t}^j, & \text{if } c_{s,j-1}^{\max} \leq t \leq c_{s,j}^{\min}, \\ \gamma_{s,j,t} \hat{y}_{s,t}^j + (1 - \gamma_{s,j,t}) \hat{y}_{s,t}^{j+1}, & \text{if } c_{s,j}^{\min} \leq t \leq c_{s,j}^{\max}, \\ \hat{y}_{s,t}^{j+1}, & \text{if } c_{s,j}^{\max} \leq t \leq c_{s,j+1}^{\min}. \end{cases} \quad (1)$$

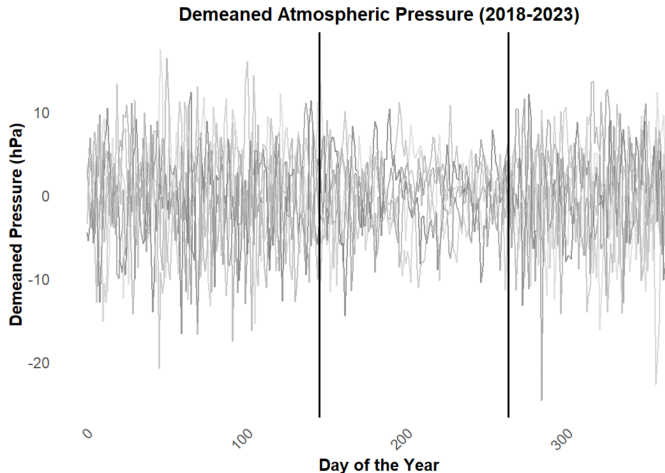
where

$$\gamma_{s,j,t} = \frac{\sum_{i \in A_t} L(i)}{\sum_{i \in \{c_{s,j}\}} L(i)}, \quad (2)$$

where $A_t := \{c_{s,j,i} \in \{c_{s,j}\} : c_{s,j,i} \leq c_{s,j,t}\}$



Multivariate change in covariance (Ryan & Killick, 2023)



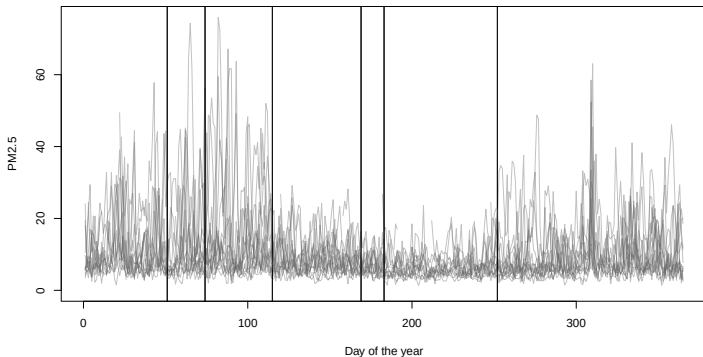
Bootstrap confidence sets using Fotopoulos et al. (2024)

	Cpt Date	Day of the year	95% C.S.
Atmospheric Pressure	May 26	146	[142, 149]
	Sep 21	264	[261, 268]

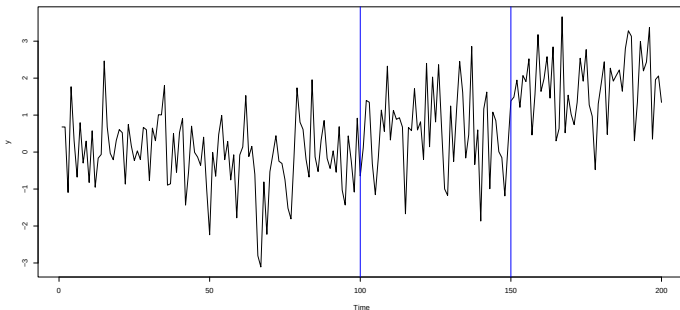
Weighted ARMA models for each segment

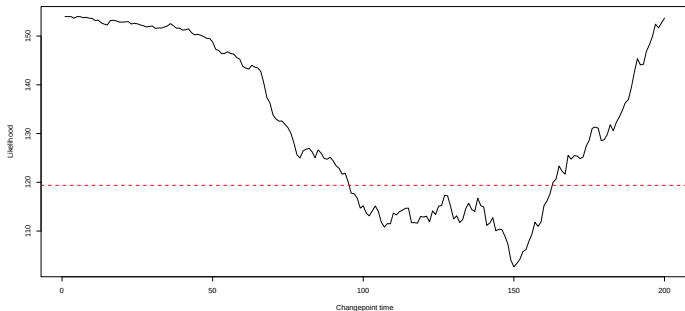
Year	MSPE			MAPE(%)		
	Blend	Cpts	ARMA	Blend	Cpts	ARMA
2018	71.18	73.09	98.63	6.61	6.83	7.89
2019	61.91	68.67	77.01	6.26	6.63	7.06
2020	66.37	78.21	x	6.38	6.59	x
2021	73.83	72.98	92.94	6.79	6.69	7.85
2022	63.31	77.76	83.58	6.07	6.93	7.31

Univariate mean+AR(1), (Beaulieu & Killick (2018)) with PELT confidence sets (Li & Killick, 2026+).

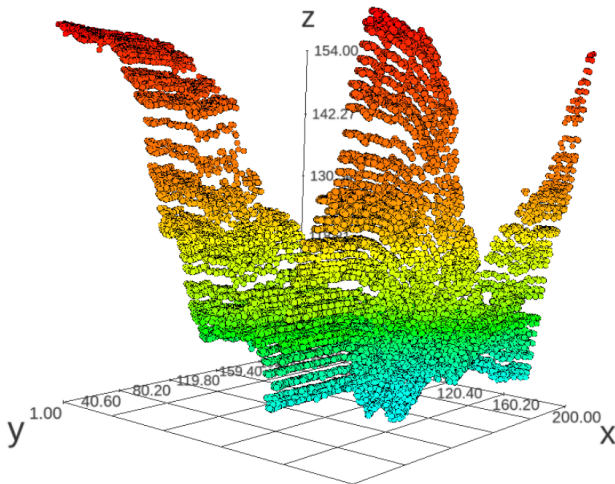


Existing confidence approaches for changepoints condiditon on the number of changes.





Two Changepoint Likelihood



- Forecast a whole new season that likely will contain changepoints
- Depicted a new measure of uncertainty in both the location and number of changepoints
- Utilize the uncertainty in the changepoint estimates to forecast
- Can be used with both univariate and multivariate changepoint approaches

Length	MSPE			MAPE(%)		
	Blend	Cpts	ARMA	Blend	Cpts	ARMA
5-years	21.86	26.77	28.3	3.05	3.57	3.8
6-years	21.83	27.18	28.7	3.05	3.6	3.84

<https://github.com/rkillick/Presentations>

Chapman & Killick (2020) An assessment of practitioners approaches to forecasting in the presence of changepoints.

<https://doi.org/10.1002/qre.2712>

Ryan & Killick (2023) Detecting changes in covariance via random matrix theory. <https://doi.org/10.1080/00401706.2023.2183261>

Fotopoulos et al. (2023) Adaptive parametric change point inference under covariance structure changes.

<https://doi.org/10.1007/s00362-023-01495-0>

Beaulieu & Killick (2018) Distinguishing trends and shifts from memory in climate data. <https://doi.org/10.1175/JCLI-D-17-0863.1>

Li & Killick (2026+) Exact Confidence Sets for Multiple Changepoint Estimation. Preprint